

Mitigating Accumulated Bias in AOA Measurements for Single-Receiver Localization Using a Calibration Source

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Abstract—Single-receiver localization using multi-snapshot angle-of-arrival (AOA) measurements offers a cost-effective solution for passive positioning. However, low-cost sensors often suffer from time-varying calibration drifts or gradual misalignments, introducing bias that accumulates over time. Ignoring this non-stationary error leads to performance deterioration as measurements accumulate, rendering standard fusion algorithms ineffective. To address this challenge, this paper proposes a joint estimation framework that models the bias as a time-linear accumulation process. A stationary calibration source at a known location is introduced to observe the bias dynamics simultaneously. By formulating a joint likelihood function involving both object and calibration measurements, the object position and the bias accumulation rate are estimated concurrently. Theoretical analysis based on the Equivalent Fisher Information Matrix (EFIM) reveals that the calibration source effectively decouples the bias parameters from the spatial geometry, thereby restoring observability. Simulation results validate that the proposed approach eliminates the bias-induced drift and achieves accuracy approaching the Cramér-Rao lower bound (CRLB).

I. INTRODUCTION

Accurate localization of non-cooperative or passive objects is fundamental to many sensing and autonomy applications, including search-and-rescue, surveillance, spectrum monitoring, and navigation of unmanned systems [1], [2]. Among several types of approaches, angle-of-arrival (AOA)-based localization is particularly attractive because it can be obtained from compact antenna arrays without stringent time synchronization and often with modest bandwidth and signaling requirements [3], [4]. When only a single receiver is available, a natural approach is to exploit receiver motion: multiple AOA snapshots collected along a trajectory can be fused to localize a stationary object, enabling single-receiver localization with minimal deployment cost [5], [6].

Single-receiver localization is appealing for low-cost deployment, yet it is generally ill-posed unless additional information is exploited. Existing solutions fall into three paradigms. Hybrid measurements combine angle with range-related observations (e.g., Time of Arrival (TOA)/Time Difference of Arrival (TDOA)) at a single receiver to resolve ambiguity [7]–[9]; e.g., [7] fuses AOA/TDOA with a closed-

form refinement, but TOA/TDOA extraction is often susceptible to phase errors, bandwidth limits, and synchronization constraints. Motion-induced diversity accumulates multiple AOA snapshots along receiver/object motion to form a virtual geometry [5], [10], yet the resulting bearings-only inference is strongly nonlinear and can be sensitive to solver stability and receiver-state uncertainties. Multipath-assisted localization exploits multipath to create virtual anchors/virtual transmitters via delay/angle estimation and association [11], [12], but typically requires resolvable multipath, wideband signaling, and nontrivial modeling/association in Non-Line-of-Sight (NLOS) scenarios. To address these constraints, this paper focuses on localizing a stationary object using AOA-only snapshots collected by a moving receiver, avoiding tight synchronization and multipath dependence. However, direct aggregation of multi-snapshot AOA becomes unreliable under time-accumulating bias.

Most existing moving-receiver AOA localization formulations assume temporally independent, zero-mean errors, or a time-invariant offset that can be calibrated offline and then treated as fixed. In practice, gradual sensor misalignment, imperfect attitude compensation, or calibration drift can induce a time-accumulating bias, making the AOA error nonstationary and correlated with the snapshot index. This bias evolution couples the bias parameter with the spatial parameters, often yielding a poorly conditioned Fisher information matrix (FIM) and degraded geometric observability. Consequently, if the accumulating bias is ignored and multi-snapshot measurements are fused as if unbiased, the localization error may increase rapidly with the number of snapshots, leading to a scenario where estimation accuracy deteriorates despite the accumulation of additional measurements. These observations motivate an explicit bias-evolution model and a calibration-aided joint estimation framework for reliable single-receiver multi-snapshot AOA localization.

To address the above challenge, this paper develops a measurement model in which the AOA bias accumulates linearly with time and is parameterized by an unknown accumulation rate. A stationary calibration object at a known location

is further incorporated and observed simultaneously, so that calibration bearings share the same bias evolution and provide direct information on the bias dynamics. The object position and the bias accumulation rate are then jointly estimated from the combined object and calibration observations. Theoretical performance limits are characterized by deriving Cramér-Rao lower bounds (CRLBs) for both object-only and joint-calibrated cases, and an equivalent Fisher information matrix (EFIM) analysis based on the Schur complement is provided to quantify the observability improvement brought by calibration. The maximum likelihood estimator (MLE) is implemented via a Gauss-Newton (GN) iteration. Simulations validate that the proposed approach significantly improves robustness against bias accumulation and achieves performance in reaching the CRLB.

The main contributions are summarized as follows:

- A single-receiver multi-snapshot AOA localization model is established where the AOA bias accumulates linearly with time and is characterized by an unknown accumulation rate.
- CRLBs are derived for object-only and joint-calibrated scenarios, and an EFIM-based analysis is provided to theoretically explain the conditioning and information gain enabled by calibration.
- A Gauss-Newton implementation of the maximum likelihood estimator is developed for joint estimation of the object position and the bias accumulation rate, and its effectiveness is demonstrated through simulations.

II. MEASUREMENT MODEL

Consider a two-dimensional localization scenario involving a stationary object, a mobile observer equipped with an AOA sensor, and a stationary calibration source. The true position of the object is denoted by $\mathbf{u}^o = [u_x^o, u_y^o]^T \in \mathbb{R}^2$. The observer starts from an initial position $\mathbf{s}^o \in \mathbb{R}^2$ and moves with a constant velocity $\mathbf{v}^o \in \mathbb{R}^2$. The calibration source is located at a known position $\mathbf{c}^o \in \mathbb{R}^2$.

The observer collects K measurements at discrete time instances $t_k = (k-1)\tau$ for $k = 1, \dots, K$, where τ represents the sampling interval. At the k -th time step, the true position of the observer is given by

$$\mathbf{s}_k^o = \mathbf{s}^o + t_k \mathbf{v}^o. \quad (1)$$

A. Object Observation Model

The noise-free bearing angle from the observer to the object at time t_k is geometrically defined as

$$\theta_k^o = \tan^{-1} \left(\frac{s_{k,y}^o - u_y^o}{s_{k,x}^o - u_x^o} \right), \quad (2)$$

where $s_{k,x}^o$ and $s_{k,y}^o$ are the Cartesian coordinates of $\mathbf{s}_k^o$. In practical scenarios, the AOA measurements are corrupted by measurement noise and a time-accumulating sensor bias. In this paper, the accumulated AOA bias over the observation window is approximated by a deterministic linear drift $t_k \beta^o$, where β^o is the unknown bias rate. This first-order

approximation is suitable for slowly varying drift over short-to-moderate durations; more general stochastic drift models, such as random walks, are beyond the scope of this paper. Consequently, the measured angle ϑ_k is modeled as:

$$\vartheta_k = \theta_k^o + t_k \beta^o + n_k, \quad (3)$$

where n_k represents the additive measurement noise, assumed to be a zero-mean Gaussian random variable with variance σ_n^2 .

Stacking the measurements for all K time steps into a vector form yields

$$\boldsymbol{\vartheta} = \boldsymbol{\theta}^o + \mathbf{t} \beta^o + \mathbf{n}, \quad (4)$$

where $\boldsymbol{\vartheta} = [\vartheta_1, \dots, \vartheta_K]^T$, $\boldsymbol{\theta}^o = [\theta_1^o, \dots, \theta_K^o]^T$, $\mathbf{t} = \boldsymbol{\kappa} \tau$ is the time vector, where $\boldsymbol{\kappa} = [0, 1, \dots, (K-1)]^T$, and $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_n)$ with covariance matrix $\mathbf{Q}_n = \sigma_n^2 \mathbf{I}_K$.

B. Calibration Observation Model

To compensate for the cumulative bias, the observer simultaneously measures the angle of the known calibration source. The true angle relative to the calibration source is

$$\phi_k^o = \tan^{-1} \left(\frac{s_{k,y}^o - c_y^o}{s_{k,x}^o - c_x^o} \right). \quad (5)$$

The calibration measurements share the same bias accumulation characteristics as the object measurements. Thus, the vector form of the calibration measurement model is expressed as

$$\boldsymbol{\varphi} = \boldsymbol{\phi}^o + \mathbf{t} \beta^o + \mathbf{e}, \quad (6)$$

where $\boldsymbol{\varphi} = [\varphi_1, \dots, \varphi_K]^T$, $\boldsymbol{\phi}^o$ is the vector of true angles computed from known positions $\mathbf{s}_k^o$ and $\mathbf{c}^o$, and $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c)$ is the measurement noise with covariance $\mathbf{Q}_c = \sigma_c^2 \mathbf{I}_K$.

The objective is to estimate the unknown parameter vector $\mathbf{m}^o = [\mathbf{u}^{oT}, \beta^o]^T \in \mathbb{R}^3$ utilizing the combined observation vector $\mathbf{z} = [\boldsymbol{\vartheta}^T, \boldsymbol{\varphi}^T]^T$, where its covariance matrix is $\mathbf{Q} = \text{blkdiag}\{\mathbf{Q}_n, \mathbf{Q}_c\}$.

III. CRAMÉR-RAO BOUND ANALYSIS

In this section, we derive the CRLB for the estimated parameters. To explicitly quantify the benefit of the calibration source, we first derive the bound using only object measurements, followed by the joint estimation case. Finally, we provide a theoretical analysis of the observability improvement based on the conditioning of FIM.

A. CRLB with Object-Only Measurements

Consider the scenario where only the object measurements $\boldsymbol{\vartheta}^o = \boldsymbol{\theta}^o + \mathbf{t} \beta^o$ are available. The unknown parameter vector remains $\mathbf{m}^o = [\mathbf{u}^{oT}, \beta^o]^T$. The measurement model is restricted to (4). The corresponding Jacobian matrix is

$$\mathbf{H}_{\text{Obj}} = \frac{\partial \boldsymbol{\vartheta}^o}{\partial \mathbf{m}^{oT}} = [\mathbf{H}_{\boldsymbol{\vartheta}, \mathbf{u}} \quad \mathbf{t}], \quad (7)$$

where $\mathbf{H}_{\boldsymbol{\vartheta}, \mathbf{u}} = \frac{\partial \boldsymbol{\vartheta}^o}{\partial \mathbf{u}^{oT}}$ contains the partial derivatives of the object bearing angles with respect to the object position. For the k -th measurement, the partial derivatives are derived as

follows. Let the squared distance be $d_k^2 = \|\mathbf{s}_k^o - \mathbf{u}^o\|^2$. The derivatives are

$$\frac{\partial \theta_k^o}{\partial u_x^o} = (s_{k,y}^o - u_y^o)/d_k^2, \quad \frac{\partial \theta_k^o}{\partial u_y^o} = -(s_{k,x}^o - u_x^o)/d_k^2. \quad (8)$$

The FIM for the object-only case, denoted as $\mathbf{J}_{\text{Obj}}$, is given by

$$\mathbf{J}_{\text{Obj}} = \mathbf{H}_{\text{Obj}}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\text{Obj}} = \begin{bmatrix} \mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}} & \mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{t} \\ \mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}} & \mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{t} \end{bmatrix}. \quad (9)$$

The CRLB is then $\text{CRLB}_{\text{Obj}}(\mathbf{m}) = \mathbf{J}_{\text{Obj}}^{-1}(\mathbf{m})$ [13]. Applying the Schur complement formula for the inverse of (9), we obtain the CRLB for $\mathbf{u}^o$

$$\text{CRLB}(\mathbf{u}^o)_{\text{Obj}} = (\mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}} - (\mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{t})(\mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{t})^{-1}(\mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}}))^{-1}. \quad (10)$$

B. CRLB with Joint Object and Calibration Measurements

When the calibration measurements $\varphi^o = \phi^o + \mathbf{t}\beta^o$ are incorporated, the Jacobian matrix of the full observation vector $\mathbf{z}^o = [\varphi^{oT}, \varphi^{oT}]^T$ with respect to $\mathbf{m}$ is

$$\mathbf{H} = \frac{\partial \mathbf{z}^o}{\partial \mathbf{m}^{oT}} = \begin{bmatrix} \frac{\partial \varphi^o}{\partial \mathbf{u}^{oT}} & \frac{\partial \varphi^o}{\partial \beta^o} \end{bmatrix} = \begin{bmatrix} \mathbf{H}_{\vartheta,\mathbf{u}} & \mathbf{t} \\ \mathbf{0}_{K \times 2} & \mathbf{t} \end{bmatrix}, \quad (11)$$

which exhibits the block structure. Utilizing the block-diagonal structure of the noise covariance $\mathbf{Q}$, the joint FIM, $\mathbf{J}_{\text{joint}}$, can be decomposed as the sum of the information from the object and the calibration source:

$$\mathbf{J}_{\text{joint}} = \mathbf{H}^T \mathbf{Q}^{-1} \mathbf{H} = \mathbf{J}_{\text{Obj}} + \mathbf{J}_{\text{Cali}}, \quad (12)$$

where $\mathbf{J}_{\text{Cali}}$ represents the information contribution from the calibration source. Since the calibration measurements are independent of the object position $\mathbf{u}^o$, $\mathbf{J}_{\text{Cali}}$ is sparse:

$$\mathbf{J}_{\text{Cali}} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & \mathbf{t}^T \mathbf{Q}_c^{-1} \mathbf{t} \end{bmatrix}, \quad (13)$$

where $\mathbf{t}^T \mathbf{Q}_c^{-1} \mathbf{t}$ is the Fisher information for the bias parameter β^o provided solely by the calibration source. Similarly, using the Schur complement, the CRLB for $\mathbf{u}^o$ in the joint case is

$$\text{CRLB}(\mathbf{u}^o)_{\text{joint}} = (\mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}} - (\mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{Q}_n^{-1} \mathbf{t})(\mathbf{t}^T (\mathbf{Q}_n^{-1} + \mathbf{Q}_c^{-1}) \mathbf{t})^{-1}(\mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{H}_{\vartheta,\mathbf{u}}))^{-1}. \quad (14)$$

Since $\mathbf{Q}_n$ and $\mathbf{Q}_c$ are definite, leading $\mathbf{Q}_n^{-1} + \mathbf{Q}_c^{-1} \succ \mathbf{Q}_n^{-1}$, it follows that $(\mathbf{t}^T (\mathbf{Q}_n^{-1} + \mathbf{Q}_c^{-1}) \mathbf{t})^{-1} \prec (\mathbf{t}^T \mathbf{Q}_n^{-1} \mathbf{t})^{-1}$. This inequality indicates that the joint CRLB is lower than the object-only CRLB.

C. Observability Analysis via EFIM

To strictly quantify the observability improvement, we analyze the Equivalent FIM (EFIM) for the position parameters $\mathbf{u}^o$. Based on the parameter vector $\mathbf{m}^o = [\mathbf{u}^{oT}, \beta^o]^T$, the FIM $\mathbf{J}_{\text{joint}}$ derived in (12) can be partitioned into a 2×2 block structure:

$$\mathbf{J}_{\text{joint}} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^T & c \end{bmatrix}, \quad (15)$$

where the blocks are $\mathbf{A} = \frac{1}{\sigma_n^2} \mathbf{H}_{\vartheta,\mathbf{u}}^T \mathbf{H}_{\vartheta,\mathbf{u}}$, $\mathbf{b} = \frac{1}{\sigma_n^2} \mathbf{H}_{\vartheta,\mathbf{u}}^T \boldsymbol{\kappa}$, $c = \frac{1}{\sigma_n^2} \|\boldsymbol{\kappa}\|^2 + \gamma \frac{1}{\sigma_c^2} \|\boldsymbol{\kappa}\|^2$, with $\gamma = 0$ for the object-only case and $\gamma = 1$ for the joint calibration case. Here, $\mathbf{A}$ represents the pure spatial information, c quantifies the information regarding the bias rate, and $\mathbf{b}$ captures the coupling between spatial geometry and temporal bias evolution.

Using the Schur complement, the EFIM for the position $\mathbf{u}^o$ is given by:

$$\mathbf{J}_e(\mathbf{u}) = \mathbf{A} - \mathbf{L}_{\text{loss}}, \quad (16)$$

where $\mathbf{L}_{\text{loss}} = \frac{1}{c} \mathbf{b} \mathbf{b}^T$ denotes the information loss induced by the uncertainty of the nuisance parameter β^o .

In the object-only scenario ($\gamma = 0$), the scalar $c_{\text{Obj}} = \frac{1}{\sigma_n^2} \|\boldsymbol{\kappa}\|^2$ is relatively small. Furthermore, under typical trajectories (e.g., linear motion), the spatial derivative columns in $\mathbf{H}_{\vartheta,\mathbf{u}}$ are often highly correlated with the time index vector $\boldsymbol{\kappa}$, making the coupling term $\mathbf{b}$ significant. This results in a large $\mathbf{L}_{\text{loss}}$, rendering $\mathbf{J}_e(\mathbf{u})$ ill-conditioned.

In the proposed joint calibration framework ($\gamma = 1$), the bias information term increases to $c_{\text{joint}} = c_{\text{Obj}} + \frac{1}{\sigma_c^2} \|\boldsymbol{\kappa}\|^2$. Since the calibration measurement noise is typically low (small σ_c^2), we have $c_{\text{joint}} > c_{\text{Obj}}$. Consequently, the subtraction term is significantly suppressed:

$$\frac{1}{c_{\text{joint}}} \mathbf{b} \mathbf{b}^T \prec \frac{1}{c_{\text{Obj}}} \mathbf{b} \mathbf{b}^T. \quad (17)$$

This inequality theoretically proves that the calibration source effectively decouples the bias estimation from the position estimation, maximizing the retention of spatial information $\mathbf{A}$.

IV. MLE IMPLEMENTED BY GAUSS-NEWTON ITERATION

Given the Gaussian nature of the noise, the MLE is equivalent to the Weighted Least Squares estimator, which minimizes the following cost function:

$$\mathcal{L}(\mathbf{m}) = (\mathbf{z} - \mathbf{z}(\mathbf{m}))^T \mathbf{Q}^{-1} (\mathbf{z} - \mathbf{z}(\mathbf{m})). \quad (18)$$

Due to the nonlinearity of $\mathbf{z}(\mathbf{m})$ with respect to the object position $\mathbf{u}$, a closed-form solution is intractable. Therefore, the GN iterative method is employed to numerically solve the optimization problem. The GN method linearizes the measurement function around the current estimate $\mathbf{m}^{(i)}$ at the i -th iteration. The update rule is given by:

$$\mathbf{m}^{(i+1)} = \mathbf{m}^{(i)} + \Delta \mathbf{m}^{(i)}, \quad (19)$$

where the update step $\Delta \mathbf{m}^{(i)}$ is the solution to the linearized normal equations:

$$\Delta \mathbf{m}^{(i)} = \left(\mathbf{H}^T(\mathbf{m}^{(i)}) \mathbf{Q}^{-1} \mathbf{H}(\mathbf{m}^{(i)}) \right)^{-1} \mathbf{H}^T(\mathbf{m}^{(i)}) \mathbf{Q}^{-1} \mathbf{r}(\mathbf{m}^{(i)}). \quad (20)$$

Here, $\mathbf{H}(\mathbf{m}^{(i)})$ is the Jacobian matrix evaluated at the current estimate as defined in (11), and $\mathbf{r}(\mathbf{m}^{(i)}) = \mathbf{z} - \mathbf{z}(\mathbf{m}^{(i)})$ is the residual vector.

The iterative process continues until the change in the cost function or the norm of the update step falls below a predefined threshold, or a maximum number of iterations is reached. The initialization $\mathbf{m}^{(0)}$ can be obtained from standard geometric intersection methods or coarse grid search to ensure convergence to the global minimum. Hence, the per-iteration

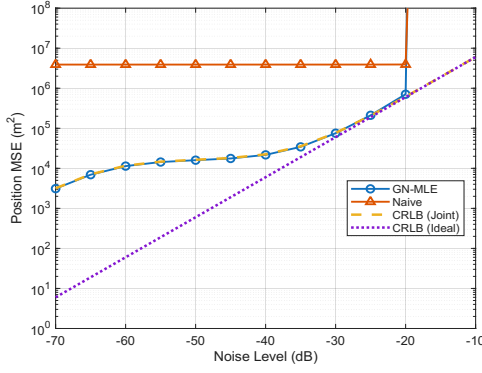


Fig. 1. Position MSE as the noise level increases.

complexity is $O(K)$, and the total complexity is $O(IK)$, where I denotes the number of GN iterations.

V. SIMULATION RESULTS

Numerical simulations are conducted to evaluate the performance of the proposed calibration-aided localization framework. The proposed Gauss-Newton Maximum Likelihood Estimator (GN-MLE) is compared with a bias-agnostic Naive estimator (which assumes $\beta^o = 0$ deg/s) and the theoretical CRLBs, CRLB (Joint) and CRLB (Ideal), derived in Section III. The simulation considers a 2D scenario with a stationary object at $\mathbf{u}^o = [3000, 5000]^T$ m and a calibration source at $\mathbf{c}^o = [-2000, 5000]^T$ m. The receiver moves from $\mathbf{s}^o = [0, 0]^T$ m with velocity $\mathbf{v}^o = [20, -10]^T$ m/s. Unless otherwise stated, the parameters are: sampling interval $\tau = 2$ s, snapshots $K = 50$. The bias rate is set to $\beta^o \approx 0.2$ deg/s. Calibration noise is fixed at a low level ($\sigma_c^2 = -30$ dB), while object measurement noise σ_n^2 varies. The performance is evaluated using the Mean Squared Error (MSE) averaged over $M = 5000$ Monte Carlo trials.

Fig. 1 depicts the position MSE against measurement noise σ_n^2 . The Naive estimator suffers from a severe error even in the small noise region, confirming that performance is dominated by accumulating bias in AOAs rather than measurement noise. In contrast, the proposed GN-MLE efficiently compensates for the bias, closely following the Joint CRLB. The results confirm that explicitly modeling the accumulating bias and jointly estimating it with the object position is crucial for robust localization. Notably, as σ_n^2 increases, the CRLB (Joint) converges to the CRLB (Ideal). This indicates that calibration measurements provide additional information on the nuisance β , reducing the performance loss due to bias uncertainty. When measurement noise dominates the error budget, the joint bound approaches the ideal bound.

Fig. 2 illustrates the MSE versus the number of snapshots K , where the object measurement noise is fixed at -40 dB. The performance of Naive estimator degrades as K increases. This occurs because the unmodeled bias drift grows linearly with time; thus, later measurements introduce larger geometric errors. Conversely, as the proposed framework correctly models such bias evolution, it converts temporal diversity into information gain and thus the GN-MLE can achieve an optimal MSE performance attaining the CRLB (Joint). This

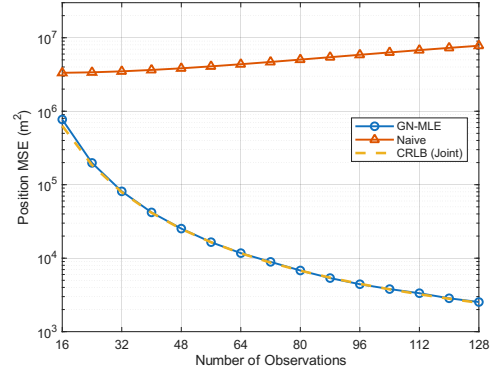


Fig. 2. Position MSE as the number of snapshots increases.

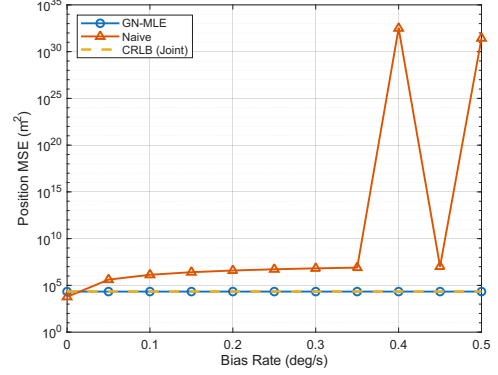


Fig. 3. Position MSE as the bias rate changes.

result underscores the necessity of the proposed model for long-duration observations.

Fig. 3 examines sensitivity to the bias rate β^o , where the object measurement noise is fixed at -40 dB. While the Naive estimator performs acceptably at $\beta^o \approx 0$ deg/s, its error escalates rapidly as the bias rate increases. The proposed GN-MLE, however, is essentially insensitive to β^o over the tested range and stays near CRLB (Joint). The result again confirms that the new measurement model for joint estimation successfully mitigates the influence from AOA bias dynamics, ensuring robust position estimation against varying accumulating bias.

VI. CONCLUSION

This work investigates the degradation of single-receiver AOA localization caused by time-accumulating sensor bias. By modeling the bias dynamics and incorporating a stationary calibration source, a joint estimation framework is developed to resolve the ambiguity between spatial geometry and temporal drift. Theoretical analysis confirms that calibration measurements significantly enhance the conditioning of the FIM by decoupling the bias rate from position parameters. Simulation results demonstrate that the proposed method is robust to bias accumulation, effectively converting long-duration measurements into localization gain rather than error drift. These findings highlight the necessity of explicit bias modeling in low-cost navigation and sensing systems. Future work will consider stochastic drift models and recursive Bayesian extensions.

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